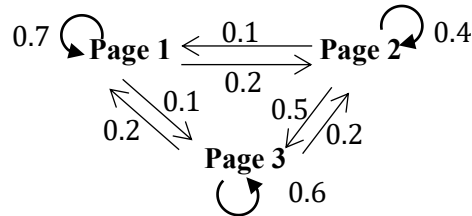


LESSON 5 Eigendecomposition and Markov Chains

Applied linear algebra is centered around two equations: the **linear system** $Ax = b$ and the **eigenvalue/eigenvector problem** $Av = \lambda v$, both of which we have learned in MATH2331. So far in this course we have focused on the linear system. In Lessons 5 to 7, we will switch to the applications of **eigenvectors** in data science. In this lesson, we will look at eigenvectors in **Markov chains**.

5.1 Markov Chains

A **finite state** Markov chain is **random process/dynamical system** with finitely many states evolving in discrete time steps. Consider a mini web with 3 pages/states.



Suppose at time $t = 0$, a surfer is on Page 1, then at $t = 1$, they are on Page 1, 2, 3 with probability 0.7, 0.2, and 0.1 respectively, and the state $x(1) = (0.7, 0.2, 0.1)$ is a **mixed state** represented by a **probability distribution vector**, where the entries sum to 1. The initial state $x(0) = (1, 0, 0)$ is a **pure state**. If $x(t) = (a, b, c)$, then

$$x(t+1) = \begin{bmatrix} 0.7a + 0.1b + 0.2c \\ 0.2a + 0.4b + 0.2c \\ 0.1a + 0.5b + 0.6c \end{bmatrix} = \begin{bmatrix} 0.7 & 0.1 & 0.2 \\ 0.2 & 0.4 & 0.2 \\ 0.1 & 0.5 & 0.6 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = Px(t)$$

To summarize: a Markov chain with states $1, \dots, n$, is characterized by a **transition matrix** P , where p_{ji} is the transition probability from state i to state j , and $x(t+1) = Px(t)$.

$$\begin{array}{c} \text{state at time } t+1 \end{array} \begin{array}{c} 1 \\ \vdots \\ n \end{array} \begin{array}{c} 1 \quad \dots \quad n \quad \text{state at time } t \\ \begin{bmatrix} p_{11} & \dots & p_{1n} \\ \vdots & \ddots & \vdots \\ p_{n1} & \dots & p_{nn} \end{bmatrix} \end{array}$$

If $x(0) = x_0$, then $x(t) = Px(t-1) = P(Px(t-2)) = \dots = P^t x_0$, and in our example,

$$x(1) = Px(0) = \begin{bmatrix} 0.7 & 0.1 & 0.2 \\ 0.2 & 0.4 & 0.2 \\ 0.1 & 0.5 & 0.6 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.2 \\ 0.1 \end{bmatrix}, \quad x(2) = Px(1) = \begin{bmatrix} 0.7 & 0.1 & 0.2 \\ 0.2 & 0.4 & 0.2 \\ 0.1 & 0.5 & 0.6 \end{bmatrix} \begin{bmatrix} 0.7 \\ 0.2 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0.53 \\ 0.24 \\ 0.23 \end{bmatrix}$$

5.2 Estimating Transition Matrix with Data

In the mini web example, we are given the transition probability matrix. In real world problems, we often have to estimate the probabilities from data! There is a simple intuitive way to learn from data in this situation. Our data is an actual long sequence of states, and

$$p_{ji} = \frac{\text{\# times the chain goes to state } j \text{ from state } i}{\text{\# times the chain is in state } i}$$

This is a **maximum likelihood estimate** (MLE) in probability and statistics. For example, if we get a sequence from a time series assumed to be a Markov chain with states 1, 2, 3:

- 3 1 1 2 1 3 2 2 3 1 1 2 1 2 3 1 2 2 3 3 1 3 1 3 3 1 2 2 3 2 1 2 1 3 2 2 3 2 2 1

then (1 1) occurs 12 times and (1 2) occurs 2 times, so $p_{11} = 2/12 = 1/6$. We will find the whole transition matrix in homework.

5.3 Eigendecomposition of Transition Matrix

We will explain how eigendecomposition sheds light on the evolution of a Markov chain. Recall that if an $n \times n$ matrix A has n linearly independent eigenvectors $v_1, \dots, v_n$, with eigenvalues $\lambda_1, \dots, \lambda_n$, then we have the eigendecomposition $A = V\Lambda V^{-1}$:

$$A = \underbrace{\begin{bmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{bmatrix}}_V \underbrace{\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}}_\Lambda \underbrace{\begin{bmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{bmatrix}^{-1}}_{V^{-1}}$$

It becomes very easy to calculate a power of A :

$$A^k = (V\Lambda V^{-1})^k = (V\Lambda V^{-1})(V\Lambda V^{-1}) \cdots (V\Lambda V^{-1}) = V\Lambda^k V^{-1}$$

where Λ^k is diagonal with $\lambda_1^k, \dots, \lambda_n^k$. This is useful for Markov chains, where we want to calculate $x(t) = P^t x_0$.

In this course we will use a calculator to find the eigenvalues and eigenvectors of a numerical matrix. In our mini web example, an eigendecomposition of P is:

$$P = V\Lambda V^{-1} = \underbrace{\begin{bmatrix} 7 & 1 & -1 \\ 5 & 0 & -3 \\ 8 & -1 & 4 \end{bmatrix}}_{\substack{v_1 \ v_2 \ v_3}} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.2 \end{bmatrix}}_\Lambda \underbrace{\left(\frac{1}{60} \begin{bmatrix} 3 & 3 & 3 \\ 44 & -36 & -16 \\ 5 & -15 & 5 \end{bmatrix} \right)}_{V^{-1}}$$

$$x(t) = P^t x_0 = V\Lambda^t V^{-1} x_0 = \begin{bmatrix} 7 & 1 & -1 \\ 5 & 0 & -3 \\ 8 & -1 & 4 \end{bmatrix} \underbrace{\begin{bmatrix} 1^t & 0 & 0 \\ 0 & 0.5^t & 0 \\ 0 & 0 & 0.2^t \end{bmatrix}}_{\substack{\text{Column picture:} \\ Ax \text{ as linear combination}}} \underbrace{\frac{1}{60} \begin{bmatrix} 3 & 3 & 3 \\ 44 & -36 & -16 \\ 5 & -15 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{V^{-1}x_0}$$

The vector $V^{-1}x_0$ has meaning. Recall that the eigenvectors form a basis for $\mathbb{R}^n$ and any vector x in $\mathbb{R}^n$ is a linear combination of $v_1, \dots, v_n$. In our example,

$$x_0 = c_1 v_1 + c_2 v_2 + c_3 v_3 = \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = Vc \Rightarrow c = V^{-1}x_0 = \begin{bmatrix} 1/20 \\ 11/15 \\ 1/12 \end{bmatrix}$$

We continue with $x(t)$:

$$\begin{aligned} x(t) = P^t x_0 &= \begin{bmatrix} 7 & 1 & -1 \\ 5 & 0 & -3 \\ 8 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1^t & 0 & 0 \\ 0 & 0.5^t & 0 \\ 0 & 0 & 0.2^t \end{bmatrix} \begin{bmatrix} 1/20 \\ 11/15 \\ 1/12 \end{bmatrix} = \begin{bmatrix} 7 & 1 & -1 \\ 5 & 0 & -3 \\ 8 & -1 & 4 \end{bmatrix} \begin{bmatrix} (1/20)1^t \\ (11/15)0.5^t \\ (1/12)0.2^t \end{bmatrix} \\ &= \frac{1}{20}(1^t) \begin{bmatrix} 7 \\ 5 \\ 8 \end{bmatrix} + \frac{11}{15}(0.5^t) \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + \frac{1}{12}(0.2^t) \begin{bmatrix} -1 \\ -3 \\ 4 \end{bmatrix} \quad \text{Column picture:} \\ &= c_1 \lambda_1^t v_1 + c_2 \lambda_2^t v_2 + c_3 \lambda_3^t v_3 \quad \text{Ax as linear combination} \end{aligned}$$

So the initial vector x_0 is “separated” into its eigenvector components and each component evolves separately. The evolved components are collected together at time t . This result can be obtained another way if we recall that $Av = \lambda v$ implies $A^t v = \lambda^t v$:

$$x(t) = P^t x_0 = P^t (c_1 v_1 + \cdots + c_3 v_3) = c_1 P^t v_1 + \cdots + c_3 P^t v_3 = c_1 \lambda_1^t v_1 + \cdots + c_3 \lambda_3^t v_3$$

From our example, we see how $V\Lambda^t V^{-1}x_0$ becomes $c_1 \lambda_1^t v_1 + c_2 \lambda_2^t v_2 + c_3 \lambda_3^t v_3$ in steps:

- First, $V^{-1}x_0$ gives c_i 's.
- Next, $\Lambda^k V^{-1}x_0 = \Lambda^k c$ brings in λ_i^k 's and we have now $c_i \lambda_i^k$'s.
- Finally, $V\Lambda^k V^{-1}x_0$ brings in v_i 's and adds up $c_i \lambda_i^k v_i$'s.

There is a similar breaking down of P^t if we do matrix multiplication as columns $\times$ rows,

$$\begin{aligned} P^t &= (V\Lambda^t)V^{-1} = \begin{bmatrix} | & | & | \\ v_1 & 0.5^t v_2 & 0.2^t v_3 \\ | & | & | \end{bmatrix} V^{-1} = \begin{bmatrix} 7 & 0.5^t & -0.2^t \\ 5 & 0 & -3(0.2^t) \\ 8 & -0.5^t & 4(0.2^t) \end{bmatrix} \frac{1}{60} \begin{bmatrix} 3 & 3 & 3 \\ 44 & -36 & -16 \\ 5 & -15 & 5 \end{bmatrix} \\ &= \frac{1}{20} \begin{bmatrix} 7 & 7 & 7 \\ 5 & 5 & 5 \\ 8 & 8 & 8 \end{bmatrix} + \frac{0.5^t}{60} \begin{bmatrix} 44 & -36 & -16 \\ 0 & 0 & 0 \\ 44 & -36 & -16 \end{bmatrix} + \frac{0.2^t}{60} \begin{bmatrix} -5 & 15 & -5 \\ -15 & 45 & -15 \\ 20 & -60 & -20 \end{bmatrix} \end{aligned}$$

This way of writing P^t as a sum of 3 rank-1 matrices, each corresponding to an eigenvector, is more illuminating than the single final expression for P^t .

5.4 Perrone-Frobenius; Stationary Distribution

In our example, we notice that both $x(t)$ and P^t converged as $t \rightarrow \infty$. This is an example of a general theorem. In matrix theory, the **Perrone-Frobenius Theorem** is of key importance for a **positive matrix**, that is, a matrix with positive entries. For the special case of transition matrices, if we order the eigenvalues in descending order (absolute value), the largest eigenvalue is $\lambda_1 = 1$ and has multiplicity 1, all the other eigenvalues have absolute value smaller than 1, and there is exactly one positive eigenvector corresponding to $\lambda_1 = 1$ with all positive entries summed to 1. This means that as time goes to infinity, the eigenvector with eigenvalue 1 dominates as $a^k \rightarrow 0$ when $|a| < 1$, and

$$\lim_{t \rightarrow \infty} x(t) = \lim_{t \rightarrow \infty} P^t x(0) = c_1 v_1$$

This distribution is also called **stationary** or **equilibrium** distribution, because

$$P(c_1 v_1) = c_1 P v_1 = c_1 \lambda_1 v_1 = c_1 v_1$$

The stationary distribution is the dominant eigenvector with elements summed to 1. A positive matrix is a special case of a **regular matrix**, which is a **nonnegative** matrix A such that A^k is positive for some integer $k > 0$. For a regular transition matrix P , there is exactly one **equilibrium** distribution such that $Px = x$. For any initial distribution $x(0)$,

$$\lim_{t \rightarrow \infty} x(t) = \lim_{t \rightarrow \infty} P^t x(0) = x_{\text{eq}}$$

The columns of $\lim_{t \rightarrow \infty} P^t$ are all x_{eq} . In our example,

$$x_{\text{eq}} = \lim_{t \rightarrow \infty} x(t) = \frac{1}{20} \begin{bmatrix} 7 \\ 5 \\ 8 \end{bmatrix}, \quad \lim_{t \rightarrow \infty} P^t = \frac{1}{20} \begin{bmatrix} 7 & 7 & 7 \\ 5 & 5 & 5 \\ 8 & 8 & 8 \end{bmatrix}$$

The stationary distribution x_{eq} gives the probability that the chain is in various states in the long run, which is the proportion of time the system spends in various states in the long run. In our example, regardless of the initial distribution, in the long run, 7/20 or 35% of surfers are on Page 1, 5/20 or 25% are on Page 2, and 8/20 or 40% are on Page 3.

To summarize, **eigendecomposition** when applied to the transition matrix of a Markov chain reveals the dominant or persistent component, in accord with one of the themes in this course:

- Matrix factorization uncovers information of a system or a data set.